

AI – POWERED LOAN ELIGIBILITY PREDICTOR

GROUP PROJECT MEMBERS

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PROJECT REPORT

COURSE – Fundamental in AI & ML

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1. Introduction

In today's time, banks receive a huge number of loan applications every day. Checking each application manually takes a lot of time and there can also be chances of human error. Because of this, many banks and financial companies are now using Artificial Intelligence and Machine Learning to make the process faster and more accurate.

This project is a simple simulation of a home loan approval system used by a bank. The system checks different financial details of a customer and predicts whether the loan should be approved or rejected.

The main idea behind this project was to understand how Machine Learning can be applied to real-life banking systems.

2. Objective of the Project

The main objectives of this project are:

- To understand how Machine Learning works in real-world applications
 - To learn how AI is used in banking and finance
 - To build a system that can predict loan eligibility using customer data
 - To improve Python programming and problem-solving skills
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3. System Overview

The system takes financial details from the user such as:

- Monthly income
- Credit score
- Loan amount
- Assets value
- Existing loan amount
- Job experience
- Number of dependents

After taking the input, the system checks some banking rules and then uses a Machine Learning model to decide whether the loan should be approved or not.

The system also includes:

- Loan eligibility prediction
 - Terms and conditions section
 - EMI calculator
 - Input validation
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4. Dataset

Since real bank datasets are confidential and not publicly available, a custom dataset was created for this project using realistic Indian banking conditions.

The dataset contains around 3000 records and includes:

Feature	Meaning
Monthly Income	Customer salary per month
Credit Score	Financial credibility of customer
Loan Amount	Amount requested
Assets Value	Total assets owned
Existing Loan Dues	Current liabilities
Job Experience	Work experience in years

The dataset was designed carefully so that the system behaves logically like a real Indian bank.

For example:

- Low income with very high loan → rejection
 - Good salary and good credit score → approval
 - Very high assets → higher approval chances
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5. Machine Learning Model

For this project, a **Decision Tree Classifier** was used.

A Decision Tree works by checking different conditions step by step and then making a final decision. It is simple to understand and works well for classification problems like loan approval prediction.

The model was trained using the generated dataset and later used to predict loan eligibility for new users.

6. Banking Rules Used

To make the system more realistic, some banking rules were added before the ML prediction.

Some important rules are:

- Minimum monthly income should be above ₹30,000
- Maximum loan limit is ₹1 Crore
- Customers with very high assets get higher approval priority
- Invalid or negative values are rejected immediately

These rules make the project closer to an actual banking system.

7. Features of the System

The project includes the following features:

Loan Eligibility Prediction

Predicts whether the customer is eligible for a loan or not.

EMI Calculator

Calculates monthly EMI based on loan amount, interest rate, and loan tenure.

Terms and Conditions

Displays bank policies and loan conditions.

Input Validation

Checks invalid inputs and prevents incorrect data from entering the system.

8. Technologies Used

The following technologies were used:

- Python
- Pandas
- Scikit-learn
- Random Module

Python concepts used:

- Variables
 - Loops
 - Conditional statements
 - Functions
 - Lists
 - Exception handling
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9. Results

The system was tested using different cases.

Some observations:

- High salary + good credit score → approved
- Low salary + large loan amount → rejected
- High assets improved approval chances
- Invalid inputs were handled properly

The system produced logical and realistic results in most cases.

10. What I Learned

While working on this project, I learned:

- Basics of Machine Learning

- How datasets affect predictions
 - Importance of logical data generation
 - How AI is used in banking systems
 - How to build a complete Python project from scratch
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11. Future Improvements

This project can be improved further by:

- Adding a graphical user interface (GUI)
 - Connecting with real-time bank databases
 - Using advanced ML models like Random Forest
 - Creating a web version using Flask or Streamlit
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12. Conclusion

This project helped me understand how Artificial Intelligence and Machine Learning can be used in practical applications like banking and loan approval systems.

Even though this is a beginner-level project, it gives a good idea of how modern financial systems use data and intelligent decision-making to reduce risk and improve efficiency.

Overall, this project improved both my Machine Learning understanding and Python programming skills.
